

DISTANCE IN KNN (OTHER DISTANCE METRICS)

Minkowski Distance– It is a generalized metric form. Depending on the value of a parameter r , it can represent the Euclidean distance (for $r = 2$) or the Manhattan distance (for $r = 1$). For other values of r , it can be used to capture various other distance concepts.

► Formula:
$$d(p, q) = \left(\sum_{i=1}^n |p_i - q_i|^r \right)^{1/r}$$

► Python Snippet:

```
from scipy.spatial import distance
dst = distance.minkowski(p, q, p=r)

#or

knn = KNeighborsClassifier(n_neighbors=3, metric='minkowski', p=... )
# You can adjust the value of p
```

PARAMETRIC VS NON-PARAMETRIC MODELS

Parametric methods make strong assumptions about the form or structure of the function that relates the input and the output ($f: X \rightarrow Y$). In other words, they assume the data follows a certain distribution or pattern.